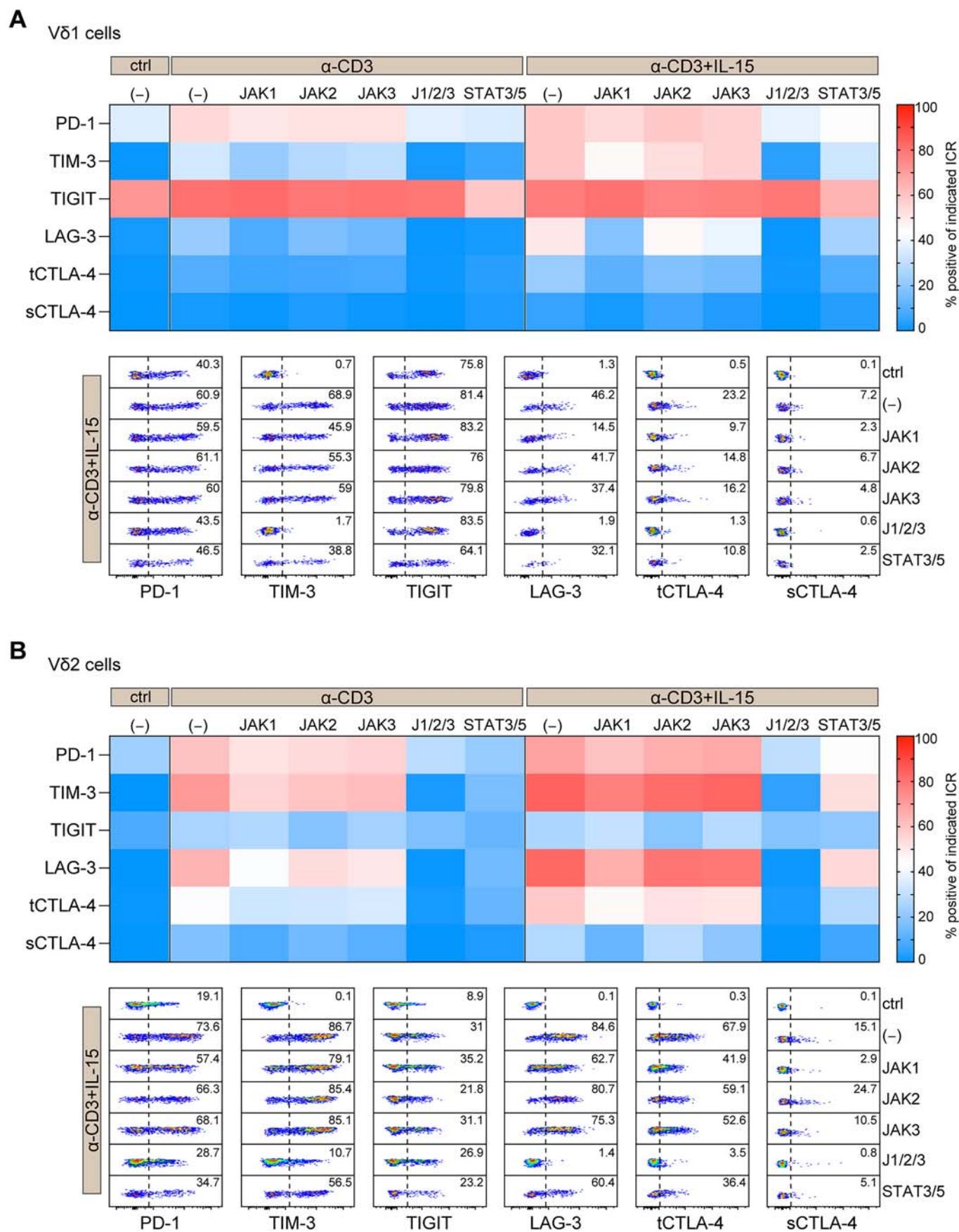


Expanded View Figures

Figure EV1. Differential regulation of ICRs in human $\gamma\delta$ T cell subsets.

Flow cytometry analysis of PD-1, TIM-3, TIGIT, LAG-3, total (t)CTLA-4 and surface (s)CTLA-4 in V δ 1 (A) and V δ 2 (B) cells. PMBCs were stimulated for 48 h with α -CD3 or α -CD3 and IL-15 or left untreated (ctrl) in the presence of inhibitors as follows: abrocitinib was used for JAK1 inhibition; NSC33994 for JAK2 inhibition; JPX-0700 for STAT3/5 dual inhibition; and tofacitinib citrate was used at a concentration 10 nM to inhibit JAK3 signaling, and at a concentration of 1 μ M for combined JAK1/2/3 inhibition. The heatmaps show average expression in cells from 13 different donors in four independent experiments. Detailed statistics for all possible comparisons in accompanying Appendix Fig. S2.



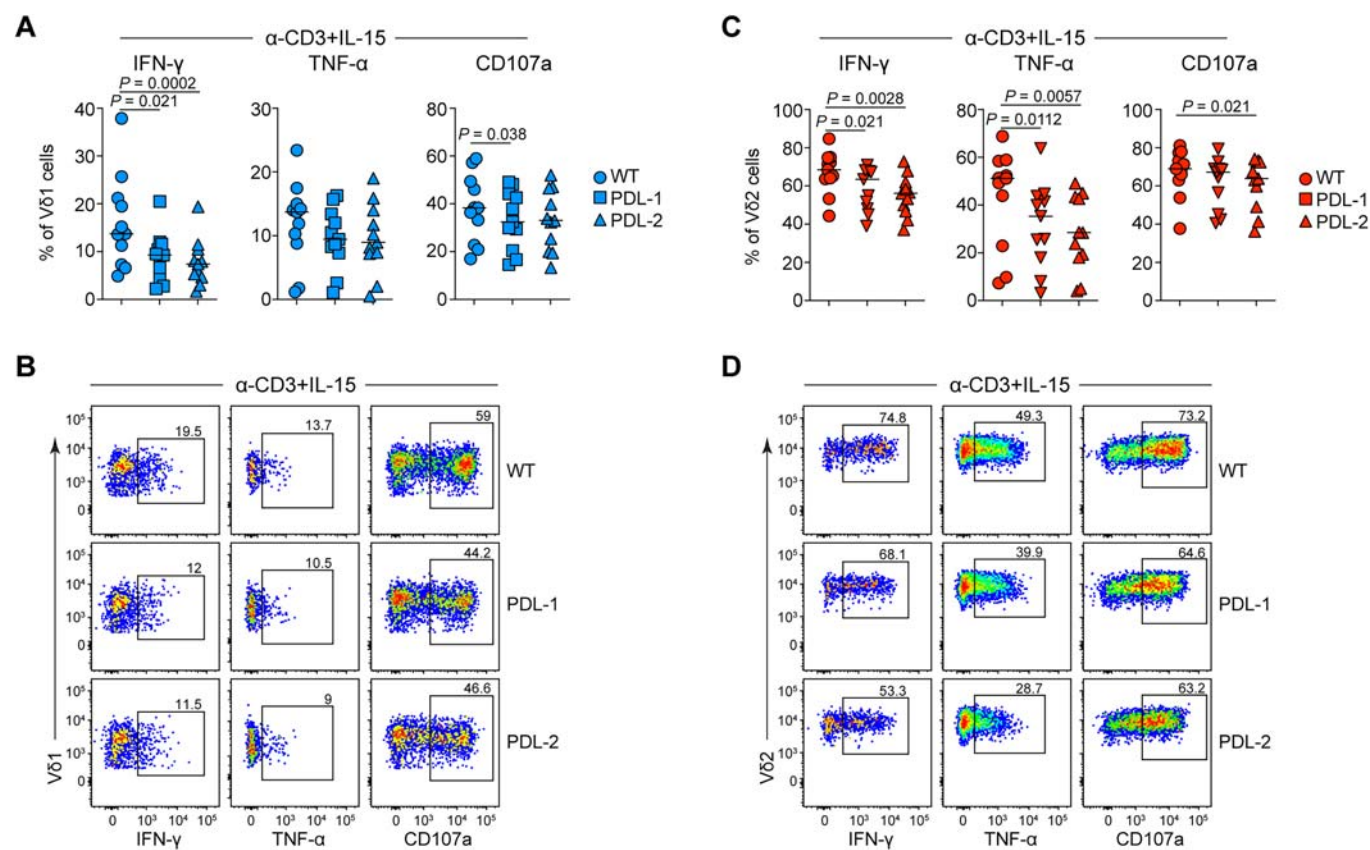
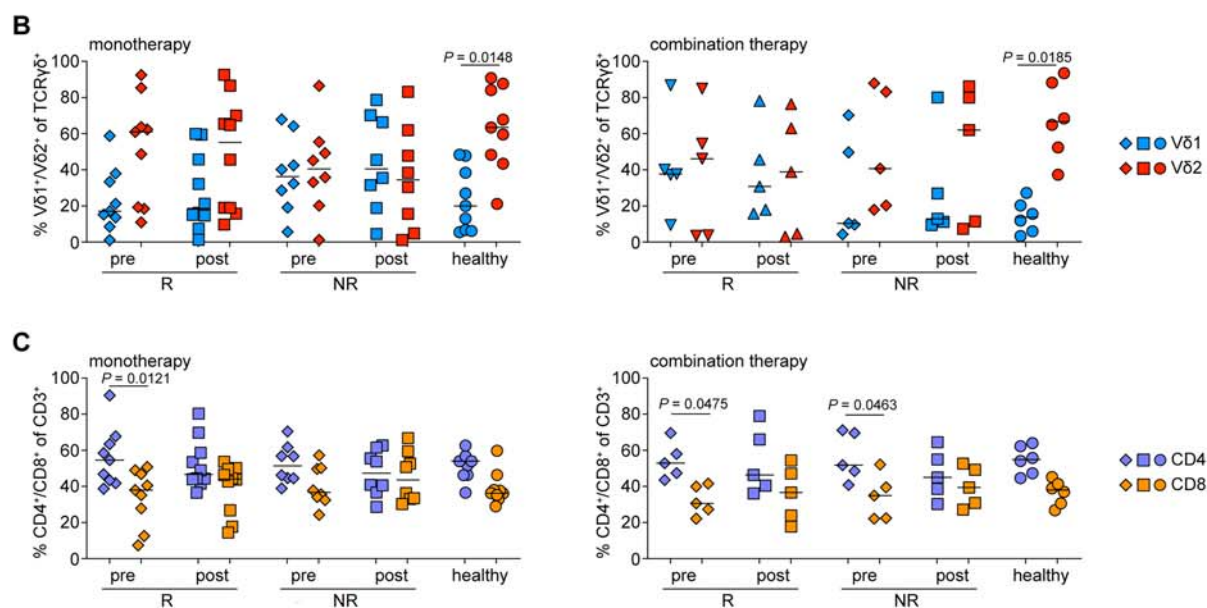
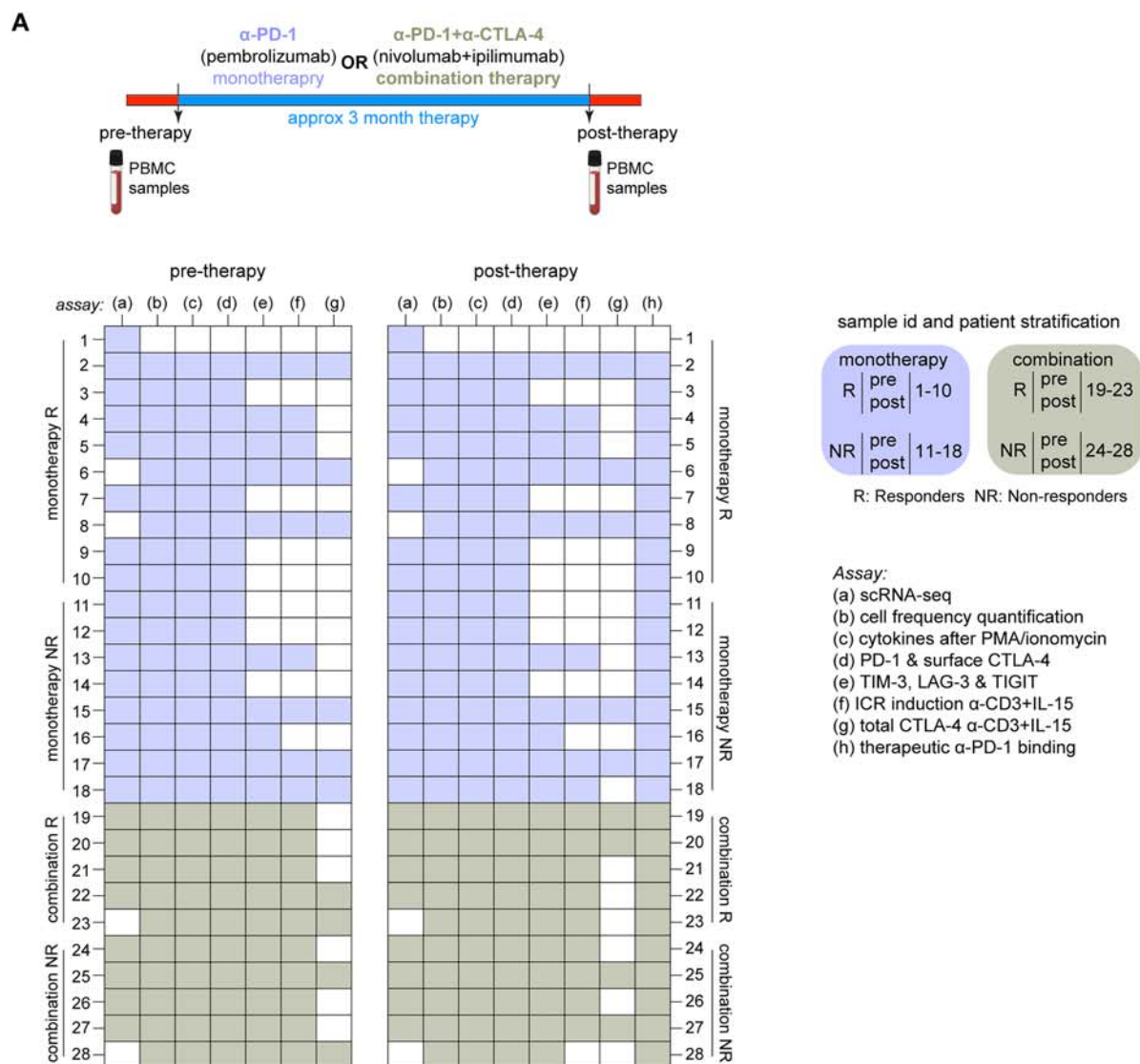


Figure EV2. PD-1 can inhibit the function of V δ 1 and V δ 2 cells.

PBMCs were magnetically depleted of CD4⁺, CD19⁺ and CD14⁺ cells and cultured for 48 h with K562 cells expressing PDL-1, PDL-2 or no ligand (WT; wild-type), in the presence of IL-15 and α -CD3 stimulation. Levels of IFN- γ , TNF- α and CD107a were assessed by flow cytometry in V δ 1 (A, B) and V δ 2 (C, D) cells. Quantification (A, C) and representative plots (B, D) from 11 healthy donors in four independent experiments. *P* values were calculated by paired Friedman test with Dunn's multiple comparisons. In graphs, each symbol represents a donor.



◀ Figure EV3. ICB therapy study design and T cell frequencies in the various patient groups.

(A) PBMC samples from patients with stage IV melanoma who were treated with either α -PD1 alone (pembrolizumab; monotherapy) or α -PD-1 and α -CTLA-4 (nivolumab + ipilimumab; combination therapy) were analyzed as shown. Samples were taken pre- and post-therapy from patients who responded (R) or did not respond (NR) to treatment. (B, C) Flow cytometry analysis depicting frequencies of V δ 1 (blue) and V δ 2 (red) cells (B) or CD4 (purple) and CD8 (orange) T cells (C) derived from patients after monotherapy (left) or after combination therapy (right). *P* values were calculated by paired two-way ANOVA and Tukey's multiple comparisons test. In graphs, each symbol represents a donor.

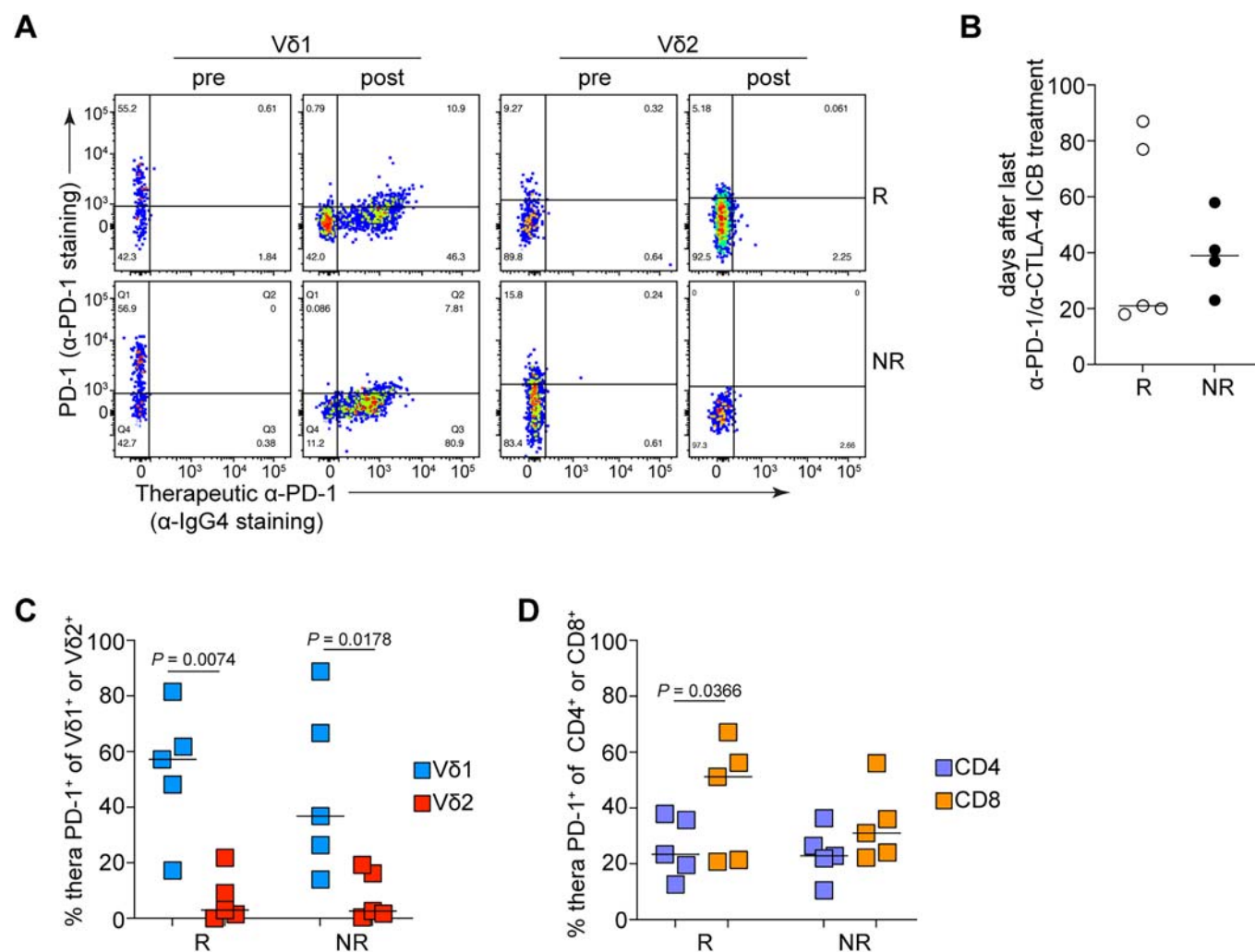
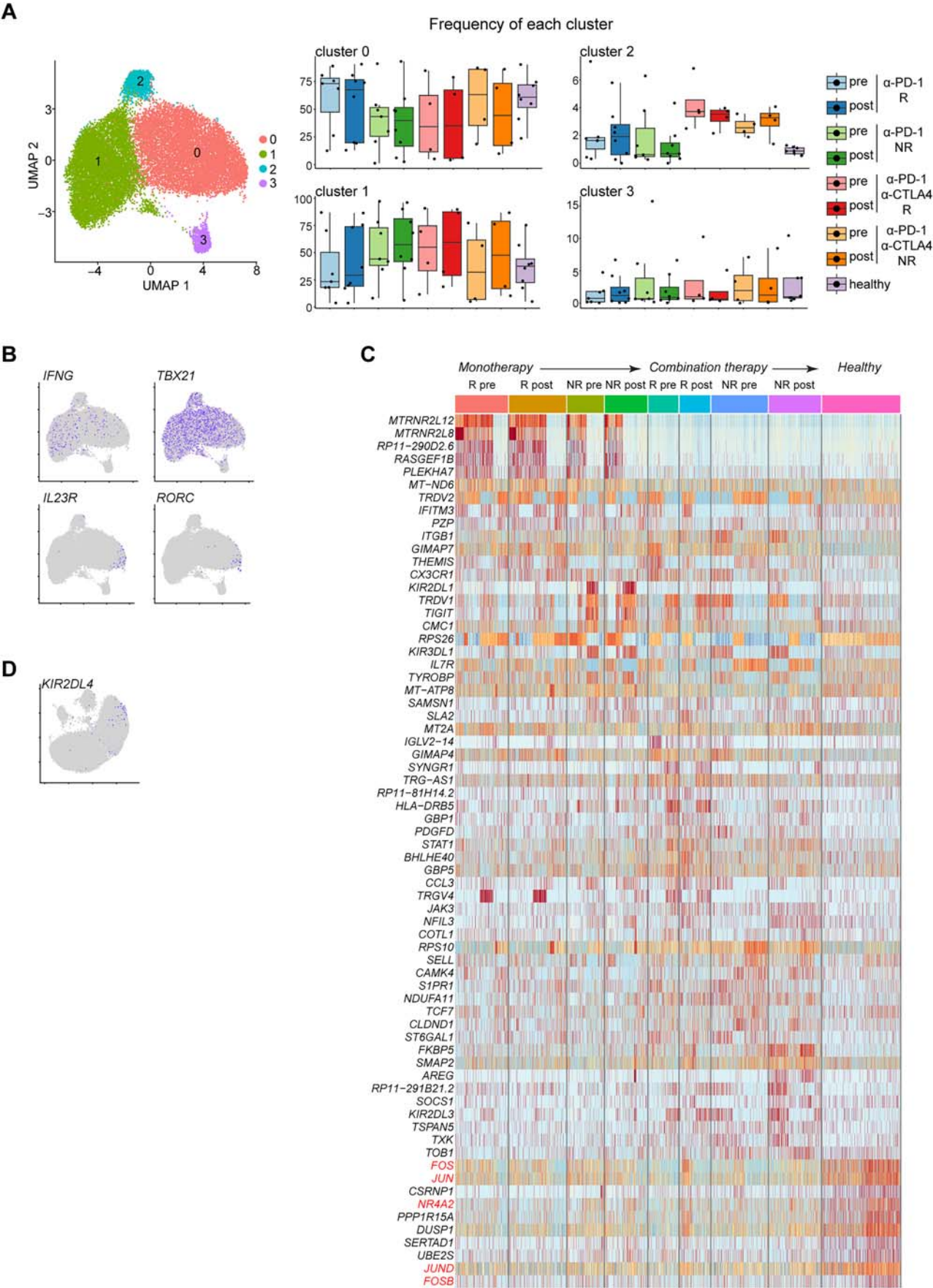


Figure EV4. Therapeutic antibody binding in patients undergoing α-PD1 and α-CTLA-4 combination therapy.

Analysis of PD-1 expression in PBMCs from patients with stage IV metastatic melanoma who either responded (R) or did not respond (NR) to a combination of nivolumab (therapeutic α-PD-1) and ipilimumab treatment. Paired samples were obtained from these patients before (pre) and 3 to 4 months after (post) the start of immunotherapy. The frequency of therapeutic antibody binding⁺ (thera PD-1⁺) cells was determined as explained in Fig. 2. (A) Representative flow cytometric analysis of therapeutic α-PD-1 antibody binding on the surface of Vδ1 and Vδ2 cells. (B) Days between the last antibody infusion and the collection of samples. (C) Thera PD-1⁺ Vδ1 and Vδ2 cells after nivolumab+ipilimumab treatment. (D) Thera PD-1⁺ CD4 and CD8 T cells after nivolumab+ipilimumab treatment. *P* values were calculated by paired two-way ANOVA and Tukey's multiple comparisons test. In graphs, each symbol represents a donor.



◀ Figure EV5. Transcriptomic analysis of circulating $\gamma\delta$ T cells from patients with melanoma and healthy donors.

(A) UMAP embedding showing clustering of circulating $\gamma\delta$ T cells ($n = 84,537$) derived from healthy donors ($n = 8$) and patients with stage IV melanoma who received α -PD-1 ($n = 16$) or a combination of α -PD-1 and α -CTLA-4 ($n = 8$) treatment (left), and the distribution of cells between clusters (right). To calculate the frequency of cells assigned to each cluster we only considered samples with at least 50 cells; each dot represents an individual sample. Each box represents the interquartile range showing 25th percentile, median and 75th percentile, and whiskers extend to the minimum and maximum values in the dataset. (B) Distribution of *IFNG*, *TBX21*, *IL23R*, and *RORC* expression in the dataset. (C) Heatmap showing normalized expression of the 10 most differentially expressed genes (DEGs) between patient groups. R, responder; NR, non-responder; pre, samples were taken before the start of treatment; post, samples were taken 3 to 4 months after the start of treatment. DEGs were identified by Wilcoxon Rank Sum test as those with an average Log2 fold change of at least 0.5 and an adjusted $P < 0.05$. (D) Distribution of *KIR2DL4* expression in the dataset.